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COMP4139

Machine Learning (MLE)

- Supervised Nonlinear Models (2)

Decision Trees & Random Forest

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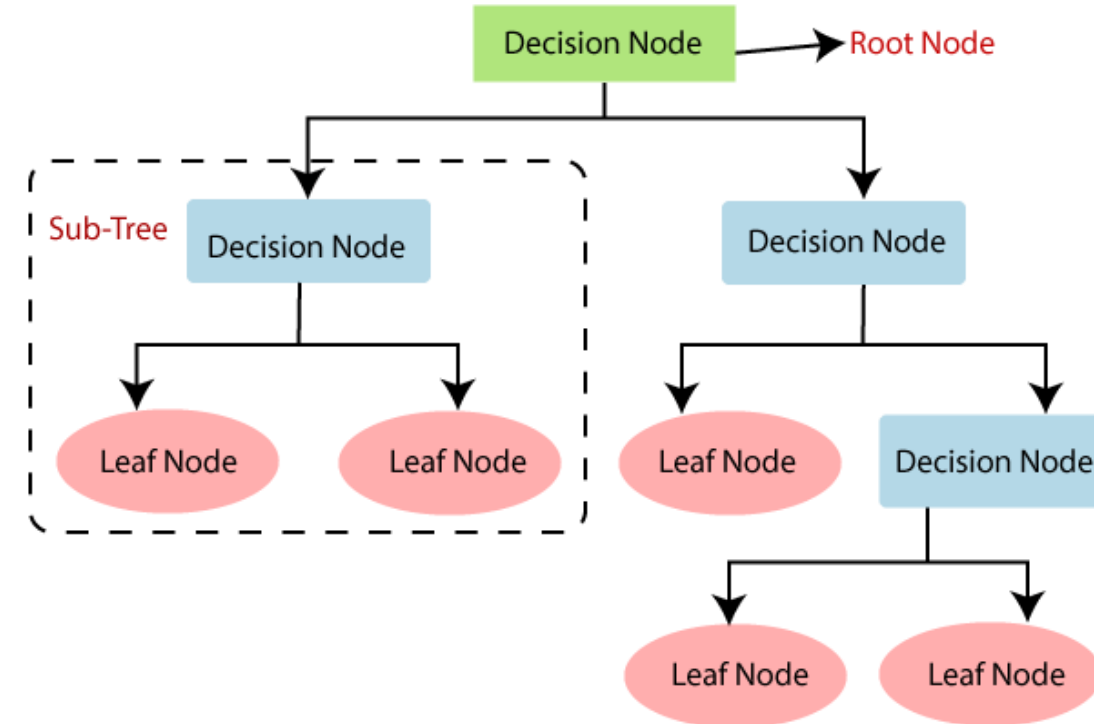
Learning Objectives

- How to build
 - Decision Tree Classification
 - Decision Tree Regression
- How (why) to extend decision tree to
 - Random Forest
- Overall Pros and Cons



Decision Trees

- Learn a “tree-like” representation of a training dataset to achieve classification/ regression of unseen data.
 - **Decision node** contains conditions to split the data
 - **Leaf node** contains the group of data and the associated label that share similar characteristics
- Many types of decision trees, but all contain similar key elements:
 - Greedy search of the best feature at each decision node
 - Criteria to select the best feature to split data in each decision node (e.g. information gain, etc.)





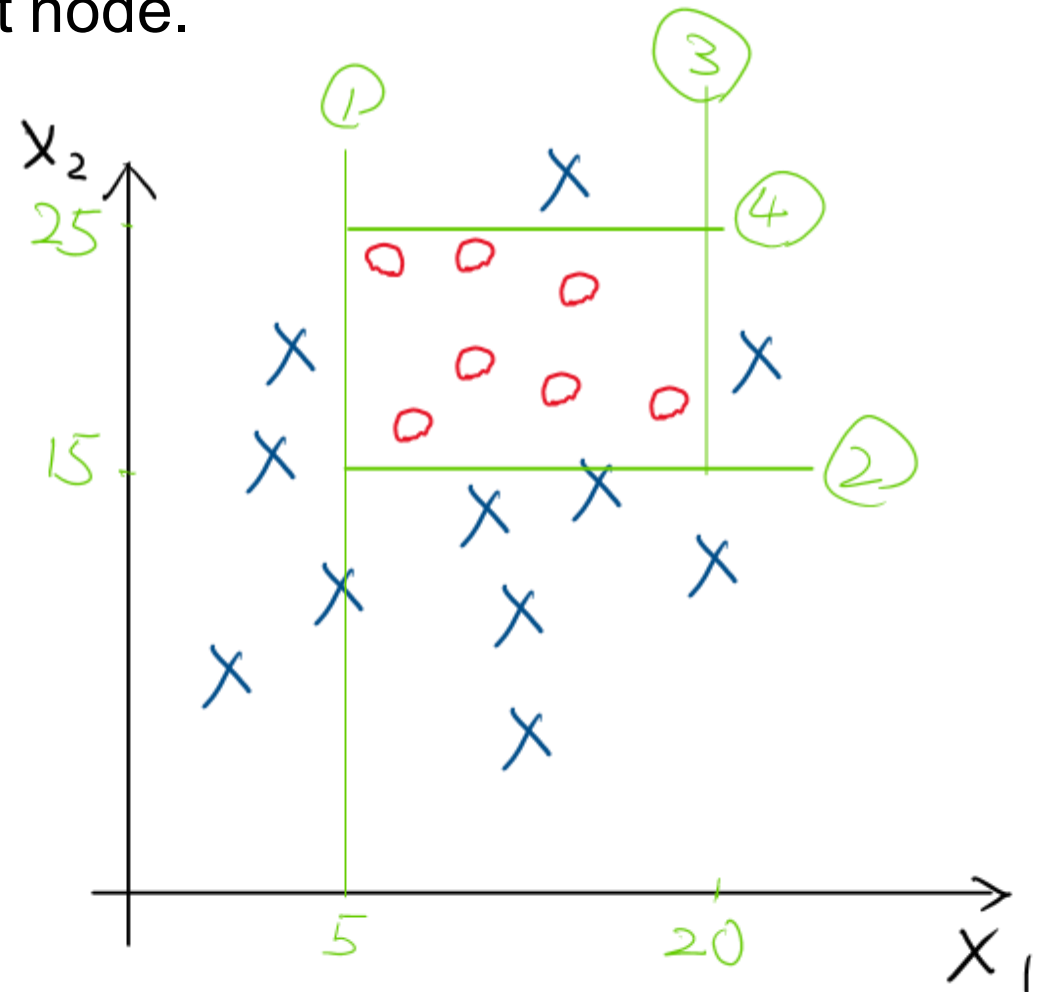
Binary Decision Trees for Classification

1. Begin the tree with the root node (R), which contains the complete dataset.
2. Find the **best feature** (attribute) using **Attribute Selection Measure (ASM)** to form a decision node
3. Divide R into subsets using the **best feature** and **best condition** (e.g. threshold).
4. For each subset find the best feature using ASM to split the data again
5. Repeat steps 2-4 until the stopping condition is satisfied (e.g. all data in the branch belong to the same class, value difference in each leaf node is small enough).



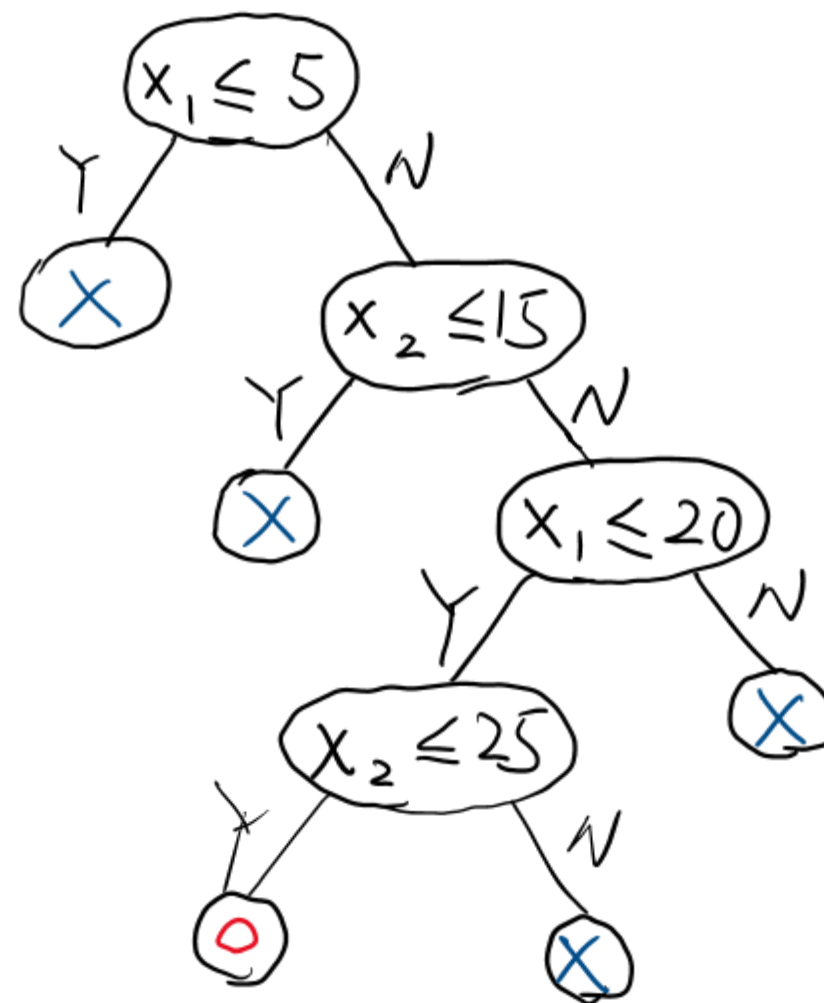
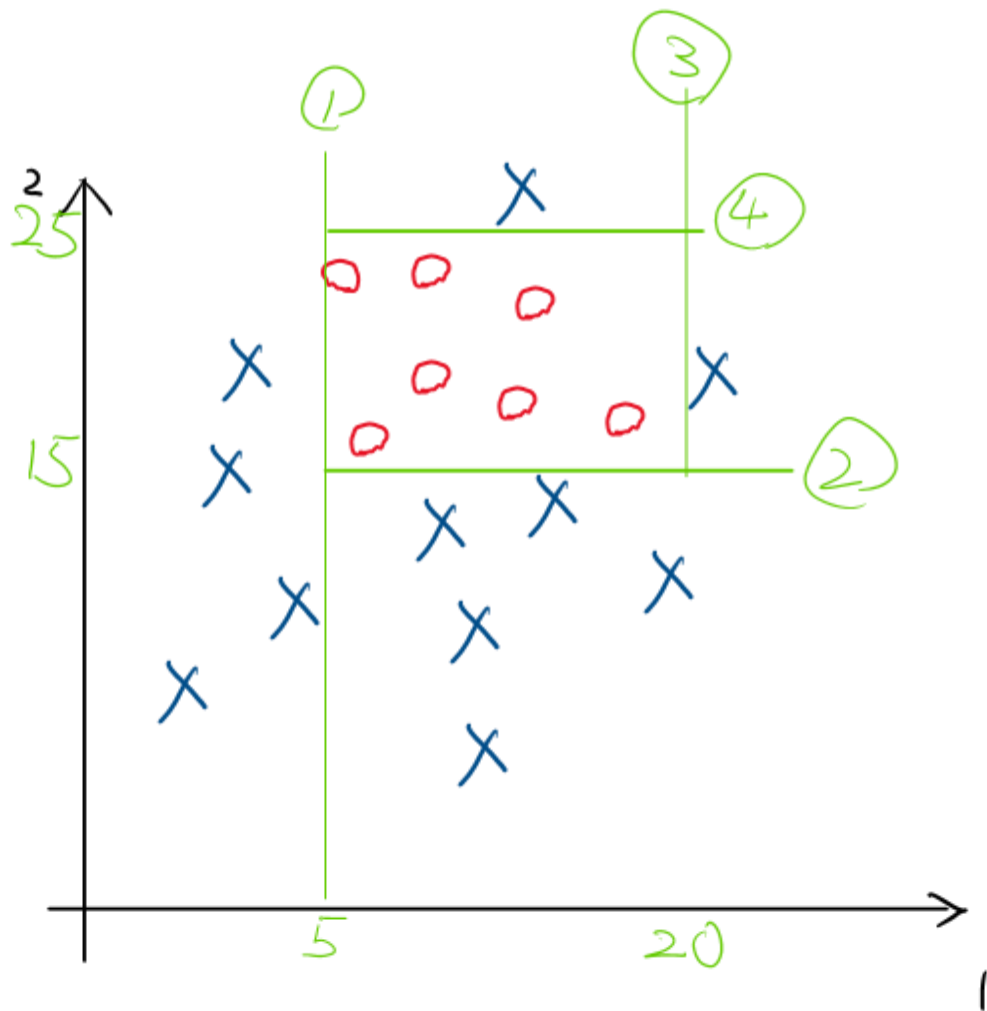
Binary Decision Trees for Classification

- Use linear decision boundaries of each feature to split the data.
- Use one feature at each node
- The same feature can be selected at different node.



Binary Decision Trees for Classification

Feature space splitting and tree structure



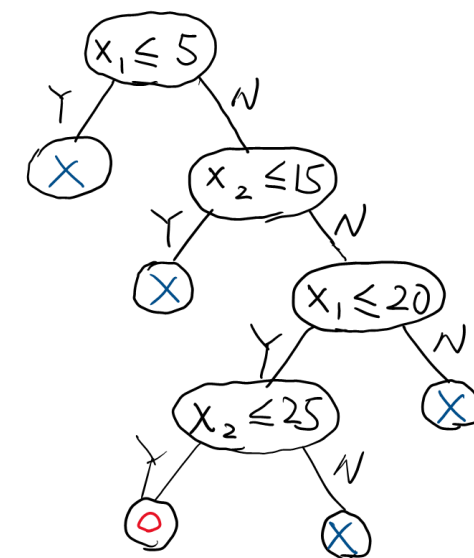
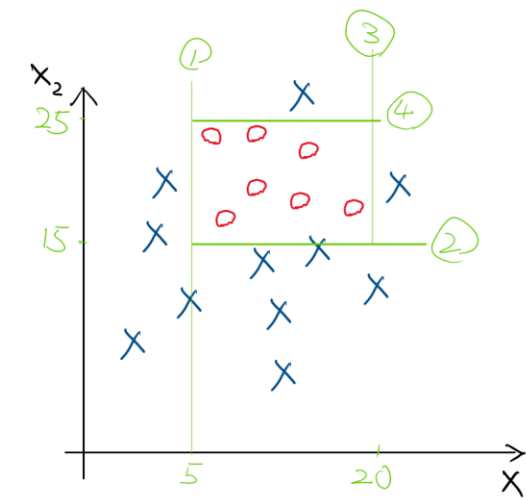
Best Feature and Condition at Each Decision Node

- Greedy Search at each decision node
 - e.g. Age feature from 10 to 100, calculate ASM using (10.5, 11.5, ..., 99.5)
- Attribute Selection Measure:
 - Entropy: measures uncertainty/purity of data S that contains C classes.

$$H(s) = - \sum_{i=1}^C P_i \log_2(P_i)$$

- Information Gain: measures the difference in entropy before and after splitting data S using feature A .

$$IG(S, A) = H(S) - \sum_{t \in T} P(t) H(t), \quad S = \bigcup_{t \in T} t$$





Example

$$H(S) = \sum_{i=1}^C -P_i \log_2(P_i) \quad IG(S,A) = H(S) - \sum_{t \in T} P(t) H(t), \quad S = \bigcup_{t \in T} t$$

Entropy at Root node: $\begin{matrix} \text{X} & 11 \\ \text{O} & 7 \end{matrix} \quad C=2$

$$H = -\frac{11}{18} \log_2\left(\frac{11}{18}\right) - \frac{7}{18} \log_2\left(\frac{7}{18}\right) = 0.9641$$

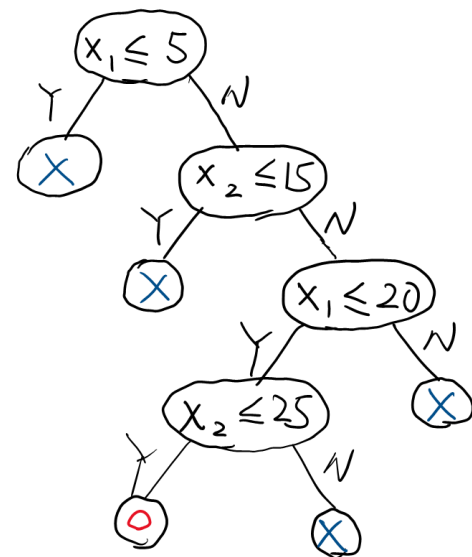
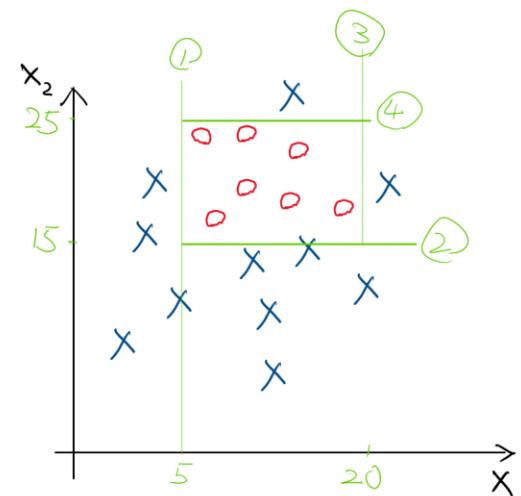
Information gain after splitting using $F_1 \leq 5$,

Left node: $\text{X}: 4$ Right node: $\text{X}: 7 \quad \text{O}: 7$

$$H(\text{left}) = -1 \log_2(1) = 0, \quad P(\text{left}) = \frac{4}{18}$$

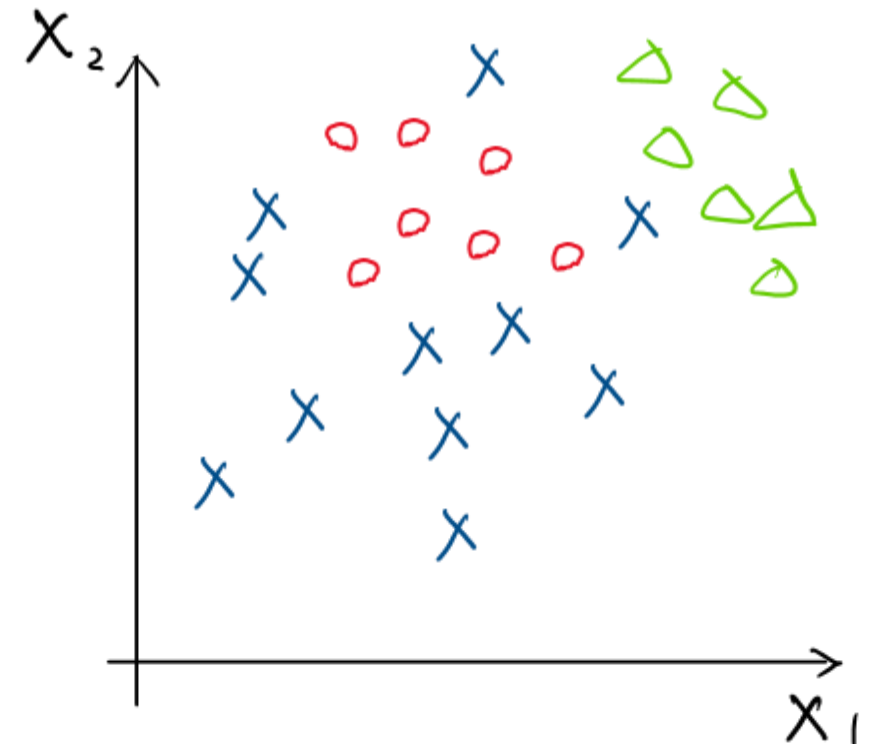
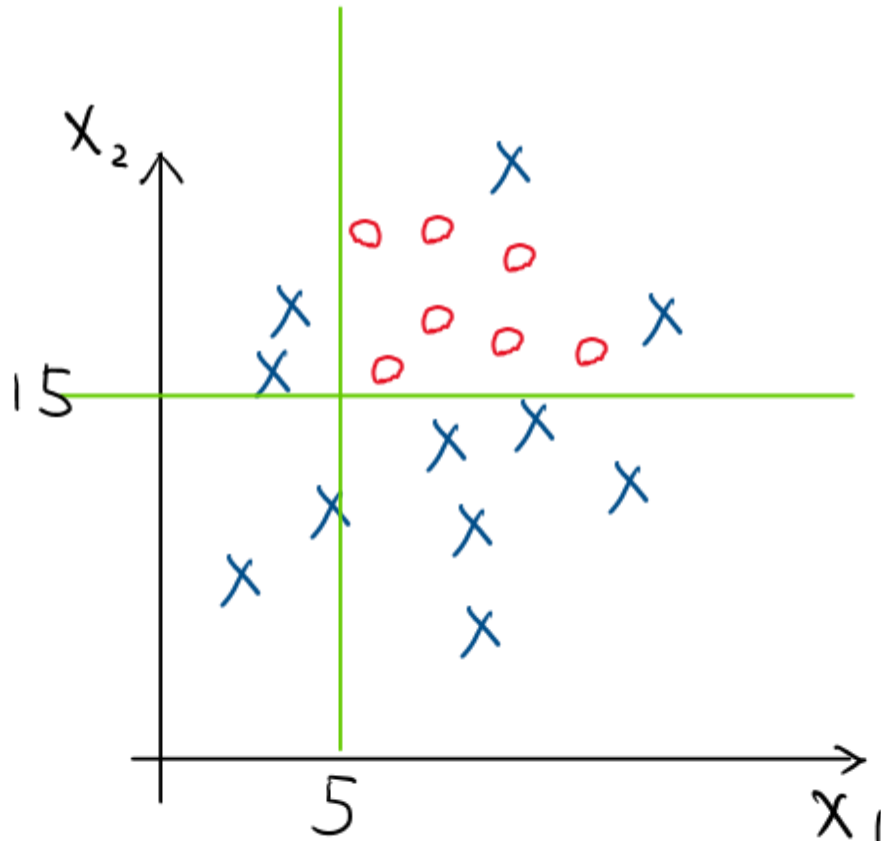
$$H(\text{right}) = -\frac{7}{14} \log_2\left(\frac{7}{14}\right) - \frac{7}{14} \log_2\left(\frac{7}{14}\right) = 1, \quad P(\text{right}) = \frac{14}{18}$$

$$IG = H - P(\text{left}) \cdot H(\text{left}) - P(\text{right}) \cdot H(\text{right}) = 0.9641 - 0 - \frac{14}{18} = 0.1863$$



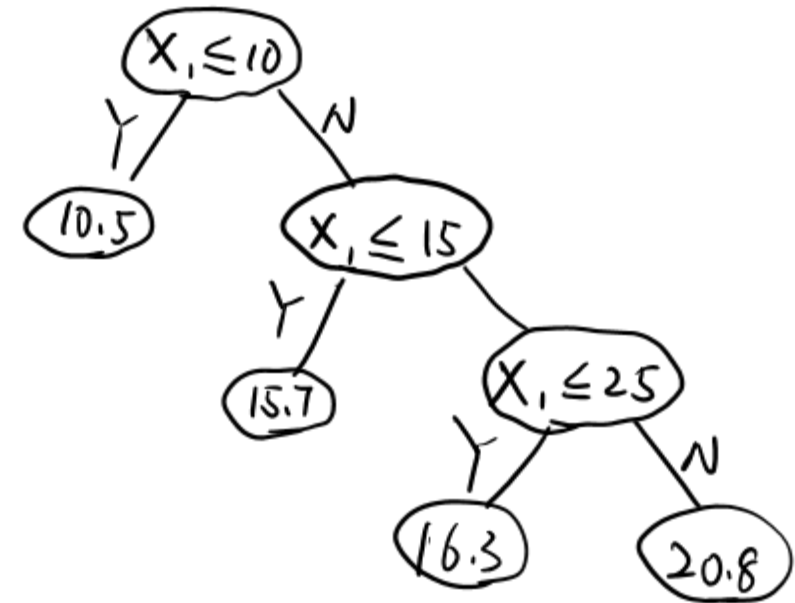
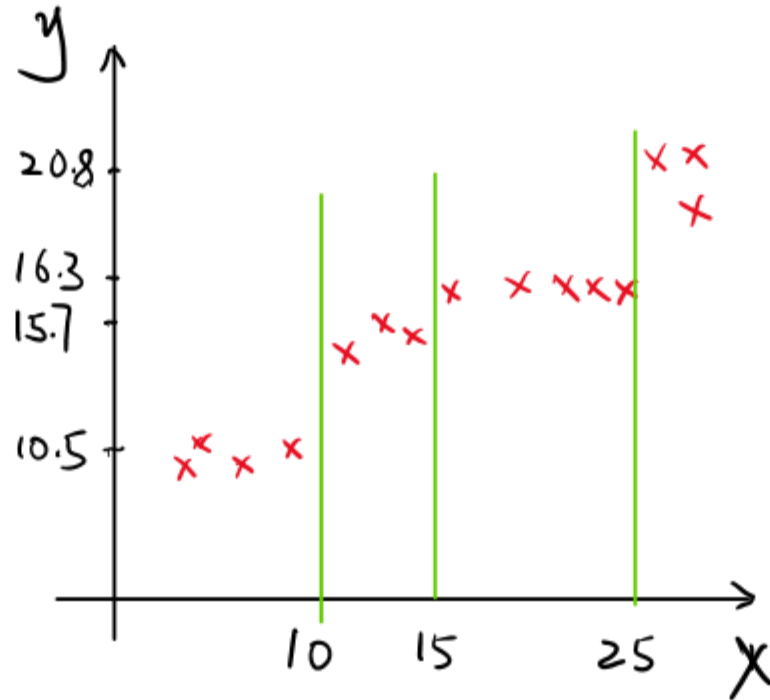
Question: Decision Tree

- Shall we use x_1 with threshold of 5 or x_2 with a threshold of 15 to split the tree?
- How to extend binary decision tree to multi-class?



Decision Tree for Regression

- Similar to classification trees, decision tree can be used for regression task.
- Use variance as the Attribute Selection Measure.



$$ASM = \sum_{i=1}^N (y_i - \bar{y}_n)^2 + \sum_{j=1}^M (y_j - \bar{y}_m)^2, \quad \text{split to } N, M \text{ subset}$$



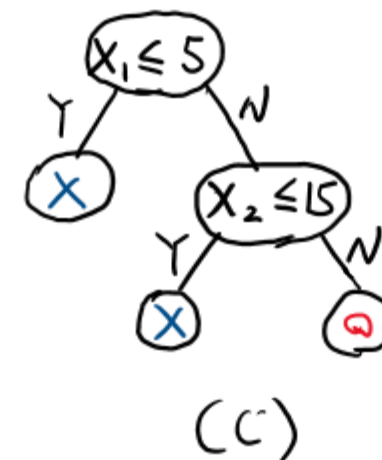
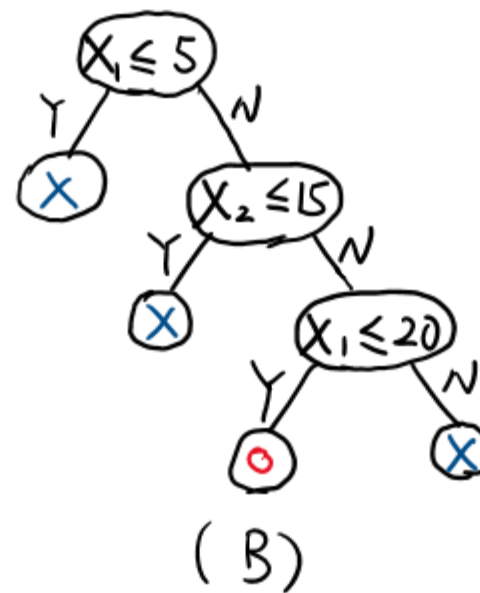
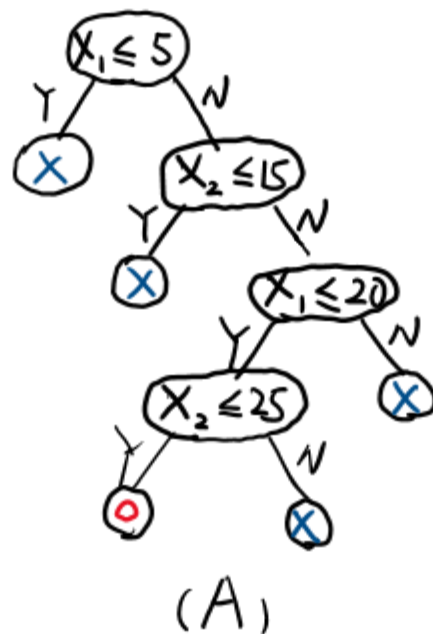
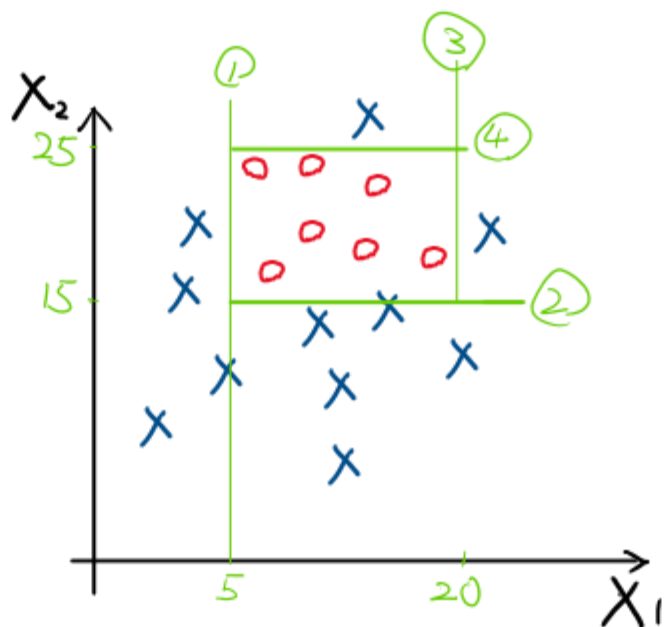
Pros and Cons of Decision Trees

- **Pros:**
 - Robust to features with different scales (don't need feature normalisation)
 - Embedded feature selection (redundant features won't be selected at decision node)
 - Scalable to handle a large training dataset
 - Could handle missing feature value by ignoring it
 - High interpretability
- **Cons:**
 - Greedy search at decision node which is computationally expensive
 - Overfitting as the tree goes deeper (biggest problem)



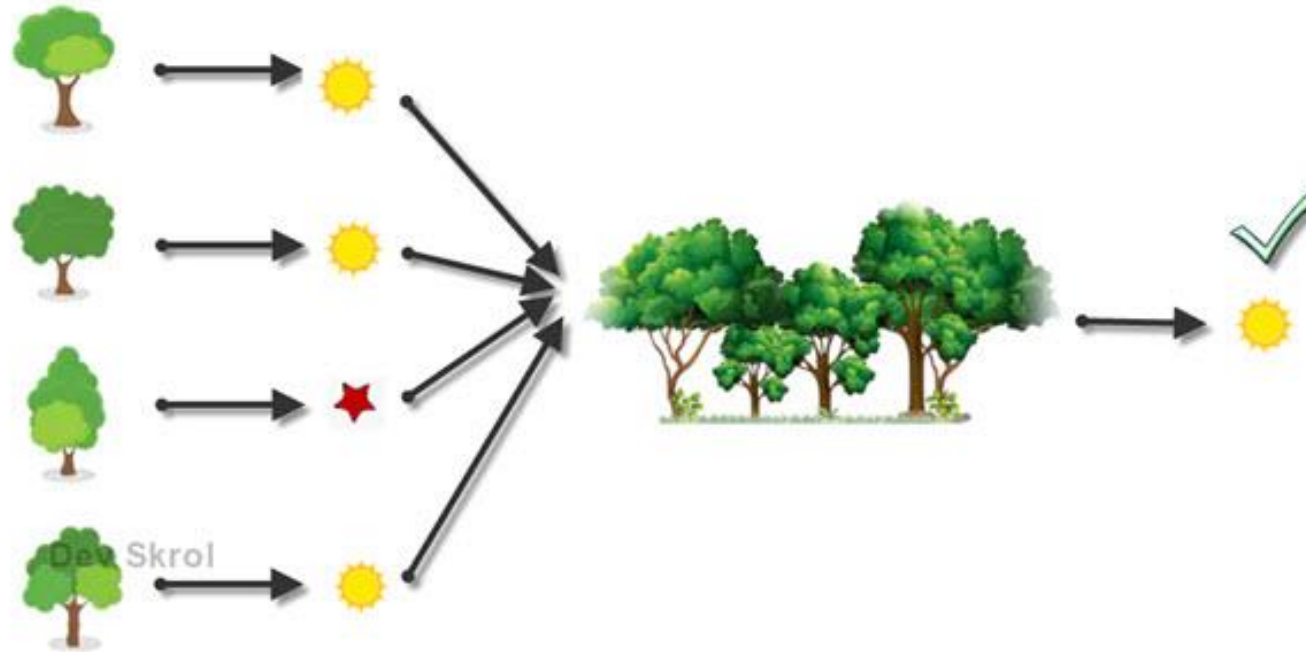
Tree Pruning to Avoid Over-fitting

Use a **validation set** to calculate error (classification/regression) for different versions of pruned trees



Random Forest

- **Ensemble method:** decision is made based on the aggregation of different **variety** of trees. How?





Random Forest

- **N samples (e.g. 1000), F features (e.g. 100), B trees (e.g. 500 trees):**
 - Final decision is based on majority voting from B decisions.
 - Regression is based on average of B trees.
 - Uncertainty can be obtained based on ratios of positive/negative votes from B trees.
- **Bagging:** the ensemble learning method that is commonly used to reduce **variance** within a noisy dataset.
 - For each of B trees, use randomly sampled 80% data for training and 20% for validation
- **Random Subspaces:** randomly sampling the features (only use a random subset of features for each tree)
 - Don't always rely on the same set of features.
 - To avoid certain features overly emphasised.
 - Classification: $F^{0.5}$, Regression: $F/3$.



Feature Importance in Random Forests

- **Permutation Importance or Mean Decrease in Accuracy (MDA):** randomly **permute feature values** of the validation samples, and the accuracy is computed each time. The decrease in accuracy as a result of this permuting is averaged over all trees. **The larger the decrease in accuracy the more important one feature is.**
 1. Pick a feature, calculate accuracy on the validation set without feature permutation.
 2. Calculate the accuracy again on the same validation set but randomly permute the feature values.
 3. Calculate the difference between the results of un-permuted and permuted version.
 4. Do this for all the features and averaged across all trees
 5. Rank the features according to this difference values: the larger the more important.



Questions: Random Forest

- Pros and cons of Random Forest compared to Decision Trees?
- What is the hyperparameters of Random Forest?
- There is another popular measure of Attribute Selection Measure: **Gini Impurity**.
Based on the same example on slide 7, replace Information Gain by Gini Impurity. (e.g. https://www.youtube.com/watch?v=_L39rN6gz7Y)



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